

CHIMBUTANE Yordine Salvador Matuele

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Education

Hong Kong Polytechnic University

Mechanical Engineering, Class of 2029.

Relevant Coursework: AI and Data Analytics, Computer Vision Classification, Embedded Systems Control.

Aga Khan Academy Maputo

Maputo, Mozambique

International Baccalaureate DP Score: 40/45. Class of 2025.

Academic Focus: Physics, Mathematics, and Economics at Higher Levels (HL).

Experiences

PolyU MATE Underwater Robotics Team | Person-In-Charge (PIC) of Buoyancy Float Team | September 2025 – Present

- Spearheaded a multidisciplinary university team in the end-to-end development, deployment, and testing of specialized underwater robotics hardware for the international MATE ROV competition.
- Engineered a custom underwater float system featuring a waterproof twist-action power mechanism and integrated depth-maintenance logic to secure 3rd place in the Hong Kong Regional competition.
- Overcame mechanical sealing challenges under high-pressure aquatic environments by seamlessly integrating physical structures with ESP32-driven embedded control systems.

PolyU Service-Learning Program Mongolia | Student Engineering Team Member | January 2026 – July 2026

- Deployed sustainable, off-grid green energy solutions in rural, underprivileged communities within Mongolia to directly tackle local power deficits and advance UN Sustainable Development Goal 7 (Affordable and Clean Energy).
- Conducted energy load calculations and configured standalone renewable hardware systems to deliver stable power infrastructure capable of withstanding extreme geographic climates.

High School Physics Research Project | Maputo, Mozambique Lead Student Researcher | September 2024 – June 2025

- Investigated the true internal resistance of commercial 9V cells by systematically varying external circuit resistance and tracking live current outputs.
- Modeled experimental laboratory data to analyze performance degradation curves across multiple consumer battery brands.

Leadership & Activities

FIRST Global Robotics Competition | Athens, Greece Mechanical Team Captain & Primary Robot Driver (Team Mozambique) | September 2024 – June 2025

- Directed structural optimization, CAD iterations, and rapid pit repairs as the Mechanical Team Captain and youngest member of the national delegation.
- Operated the competition robot as the primary driver during high-pressure international tournament matches against global teams.

Competitive Football Club | Maputo, Mozambique Varsity Sports Captain | September 2023 – June 2025

- Cultivated team cohesion, strategy development, and athletic discipline while leading a highly competitive squad through regional championship fixtures.

Skills & Interests

- **Technical:** Solidworks (Proficient), Fusion 360 (Proficient), Python Programming (Intermediate), C++ / Arduino Development (Intermediate), Prompt Engineering, Web design (Intermediate).
- **Language:** English (Fluent/Native), Portuguese (Fluent/Native), Mandarin (Learning).
- **Laboratory:** LoRa Module Hardware Integration, ESP32 Microcontrollers, Circuit Load Testing, Laser Cutting & 3D Printing Prototyping.
- **Interests:** Competitive football, sustainable marine energy systems, space-grade computing infrastructure (orbital data centers), and autonomous robotics.